

Privacy 101

Introductions

I'm just a third year CS student that's very passionate about free software and digital sovereignty

Not some super hacker or a giga privacy nerd

Why I'm Holding This Workshop

Because There's usually 3 categories of people that care about privacy:

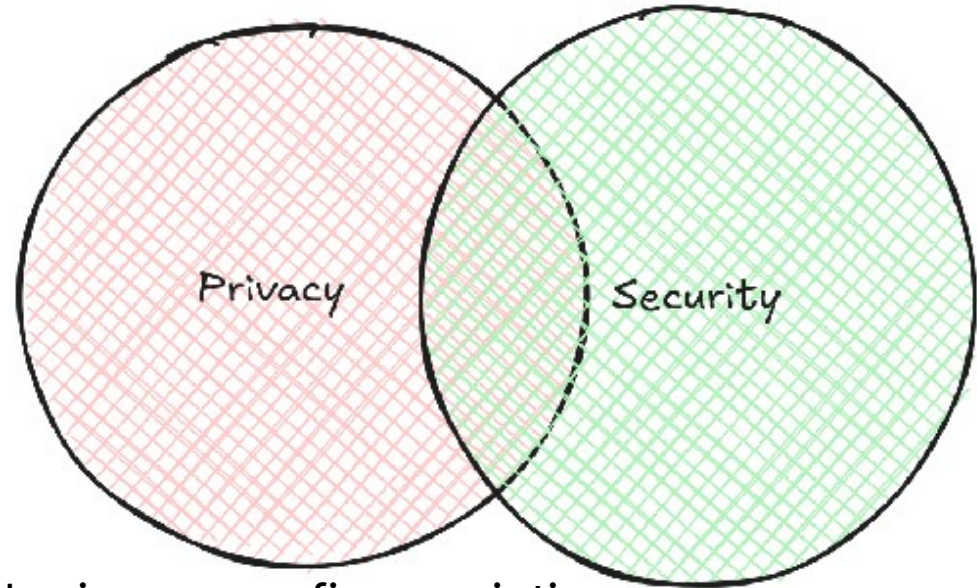
- The nerds
- The paranoid (there is some overlap with these 2)
- People who had their privacy violated

Many times that last group of people don't even have the tools to understand what happened to them.

Disclaimers

Privacy not Security

- Todo add better venn diagram
- Privacy and Security are 2 different things, but they can intersect



This guide wont help you against OSINT, finacial privacy, nor fingerprinting

I cut more than 70% of the material

This is the list of some of the things I could talk about

Scaletta

misc:
you follow the rules of something you don't believe in (video di zay duplee di carlo)

- Introduction

- Disclaimer

- Privacy and security are different things even tho they can overlap (show a venn diagram)
- This guide wont help you against OSINT either
- this guide doesn't help with financial privacy, if you want that use cash and support the local economy
- Most of the stuff I wanted to talk about were cut to make everything more streamlined, I would love to talk to you after the workshop, I will answer everything that comes my way

- Why

- Information is power, power being held by something whose best interest goes against yours (company will do everything to monetize you)
- " ubihhh but I've got nothing to hide" (sigh)
- Instagram sending makeup ads when a girl's self esteem was at their lowest (when she deleted a photo of herself from the account)
- food prices based on behaviour patterns (video mandato a carlo)
- Recent aggrgious data usage, (pokemon go and niantic using the data , windows, meta) / Worst offenders / false positive cases cases → using childer and security as an excuse for privacy invasion
- Some companies lie about it (apple)
- shifting frame fallacy (crabs being slowly boiled, explanation in samurai wallet interview) → comparisons from long ago and now
- If a human were to do it it would be creepy, so why its okay for an algorithm to do it
- The tech monopolies (various google services etc)
- Governments are in it too (palantir, persona endpoint fiasco, cia global espionage projects, poste italiane che chiedono ai developer di tor di aggiungere una backdoor)
- People care about privacy only after it has been violated, and many times they don't have the tools to understand what happened to them.

- How

- Advertising IDs
- Cookies, what are they
- fingerprinting (and how this guide doesn't really help with it)
- Analytics

- Non Identifiable data and how they identify you anyways, (Non identifiable data doesn't exist)

- What can you do about it

- Email(proton/tutanota), ypns (proton)
- Privacy respecting alternatives
- Self hosted services
- Alternative front ends
- Open source operating systems
- ~~Piracy~~

- The sad truth,

- you might struggle at first,
- u need to be less convinient (reject convenience),
- some stuff cant be made privacy proof, sometimes its easier to stop using something than to make it private
- social media is evil and there's little you can do about it
- Threat models and Perfect is the enemy of good

- Bad Services:

- Incogni/Delete.me are not that great(reject convenience video)
- Ring/other online video camera (data breaches, recent dog feature)
- VPNs (spread disinformation with ads)
- DNA services (23andMe databreach)

- AI

- Italian privacy thing (multa ad openAI)
- Use local models whenever possible (self hosting call back)
- If you cant use duck.ai, because its an aggregator
- What you type will end up in the dataset and its not averaged, your personal details could end up in the reply of someone's asnwser (artists works being copied 1 to 1 in img to img etc)
- AI is dangerous in so many other ways but this is not the time to talk about it so ask me questions later if you want

- A word about the future / keep up with the everchanging times

- learn english (the world you will live in is being made overseas and Italy's coverage of these topics is surface level at best)
- keep yourself informed
- follow social media channels that keep you informed (lewis rossman,
- recent age verification (identity verification) bills → we are building a framework to determine who has access to information and who doesnt
- Please talk about this, this is not just nerd stuff, this is a human right being destroyed while everyone's watching
- there's a war on privacy and those that make privacy tools (france and graphene os, age restriction laws at the OS level,
- **Ending**
- Made with 100% Free software
- Useful tools (privacy visualizer, consumer rights wiki
- Great places (reject convenience, mental outlaw, New Design Congress, Internet of Bugs, Ai now institute, Naomi Brockwell TV)
- Recommended reads (The circle, interview with signal ceo, who will remember us when the servers go dark)
- Only self hosted ai was used in the making of this workshop

I've had to remove lots of stuff to avoid fearmongering

Why should you care?

Everyone should care about privacy

- Its a universal right

No one shall be subjected to arbitrary interference with their privacy, family, home or correspondence, nor to attacks upon his honor and reputation. Everyone has the right to the protection of the law against such interference or attacks.

— [Universal Declaration of Human Rights, Article 12](#)

"the principle which protects personal writings and any other productions of the intellect or the emotions, is the right to privacy."
- Samuel D. Warren II and Louis Brandeis, 1890

This information can be used to harm you

- With enough data a company can know you better than yourself, this can lead to emotional exploitation, false positives, economic discrimination and more.
- Many companies have bad security policies
- Data breaches can be devastating for identity theft or automated AI phishing campaigns

A couple of examples

Whistleblower: Meta sold ads when teens felt “worthless”

“It could identify when they were feeling worthless or helpless or like a failure, and [Meta] would take that information and share it with advertisers,” Wynn-Williams [stated](#) to the Judiciary Committee’s subcommittee for crime and terrorism. “Advertisers understand that when people don’t feel good about themselves, it’s often a good time to pitch a product — people are more likely to buy something.”

She elaborated that Meta informed advertisers about teens’ depressed states to optimize ad timing. For example, [a teen girl deleting a selfie might trigger ads for beauty products](#), or concerns about body confidence could lead to weight loss ads.

A dad took photos of his naked toddler for the doctor Google flagged him as a criminal

Mark did not remember this video and no longer had access to it, but he said it sounded like a private moment he would have been inspired to capture, not realizing it would ever be viewed or judged by anyone else.

“I can imagine it. We woke up one morning. It was a beautiful day with my wife and son, and I wanted to record the moment,” Mark said. “If only we slept with pajamas on, this all could have been avoided.”

And More

Your Grocery Store is About to Start Charging You More for Being Poor, Old, or Shopping at the Wrong Time

‘Traumatizing’: Woman Warns Against Ring Cameras, Says Man Started Talking To Her Toddler In The Middle Of The Night

“But I have nothing to hide”

Privacy is not the right to hide, it's the right to chose who our data is shared with

Good news

You can do something about this

But First

- Before all that, you don't need to do everything that I'm going to tell you, some stuff might be too technical for you to achieve or you might not be able to stop using privacy invading services for work reasons.
- That's why I'm going to borrow a term used in cyber security: Threat Modeling

What is a Threat Model?

- It is the process of defining your "Privacy Budget." It is a balance between Security, Convenience, and Effort.
- Your "Threat Model" depends entirely on what you are afraid of:
 - If you just want to avoid annoying ads, you just need a better browser.
 - If you are a journalist protecting a source, you need encrypted messaging and specialized operating systems.
- To build one, ask yourself these three questions:
- What is my most valuable data?
- Who am I protecting my data from?
- How much "friction" can I live with?

Always remember: **Perfect is the enemy of good.**

Start small and simple and build on top of what you've already done if you feel it necessary.

What is your data?

- Personally Identifiable Information
- Non Personally Identifiable information
- They can also be referred to as Direct and Indirect information

PII stands for “personally identifiable information,” and it refers to any information that can identify or trace an individual either directly (direct identifiers) or when paired with other information (indirect identifiers). Here are some examples of each:

Direct Identifiers

- Name
- Address
- Social Security number (SSN)
- Other identifying number or code
- Phone number
- Email address

Indirect Identifiers

- Gender
- Race
- Birthdate
- Geographic indicators

Non Identifiable Information Is a Lie

- Given enough data points any device can be identified
- Companies claim they only collect "anonymous, indirect data" to avoid legal trouble.

Advertising IDs, Cookies and Trackers

- Cookies: Digital breadcrumbs left inside your browser to remember who you are on a specific website.
- Trackers: Invisible scripts embedded in websites and apps that report your activity back to third parties (like Meta or Google).
- Advertising IDs: A unique, permanent code assigned to your physical phone and accounts.

Data Brokers

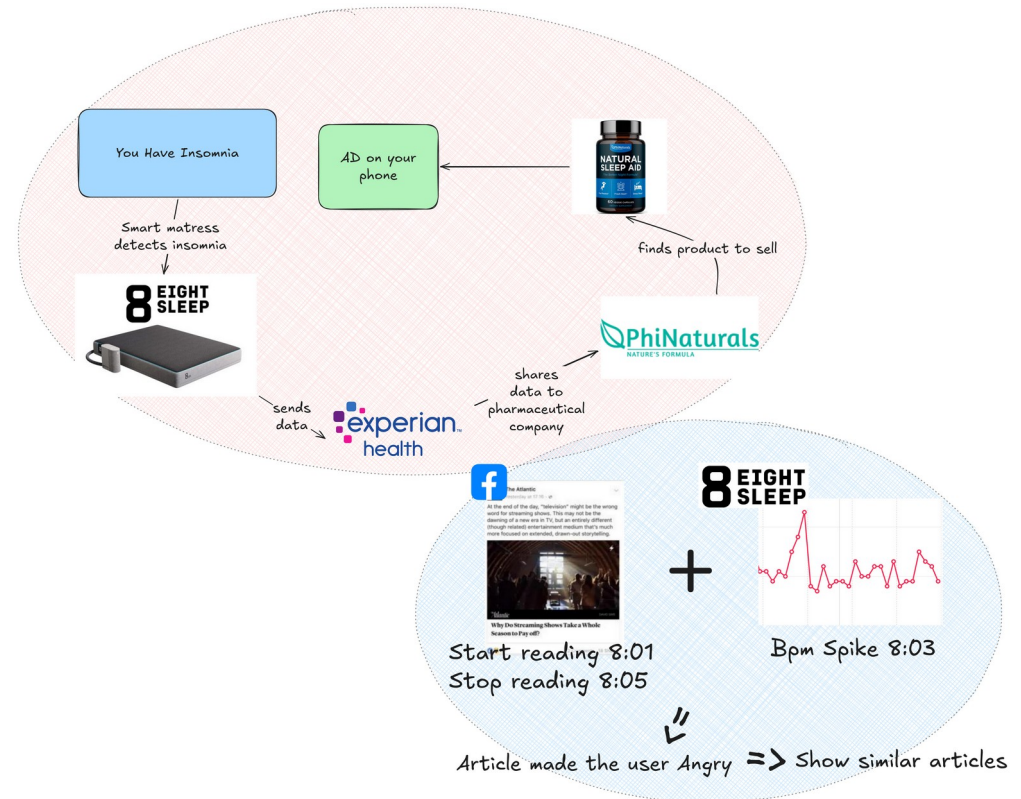
- Multi-billion dollar corporations you have never heard of, collecting data on people who never gave them permission.
- They purchase your online tracking data and merge it with offline public records (voter registries, property deeds, store loyalty cards).
- Your profile is packaged into categories (e.g., "Socioeconomic Level 3," "Expectant Mother") and sold to the highest bidder.

4 Major categories:

- Personal Health
- Financial Information
- Risk Mitigation
- Marketing

Personal Health Data Brokers

- They collect an obscene amount of sensitive information.
- They collect:
 - Health data
 - Usage data of tools
 - Search History
 - Purchase History



Risk Mitigation Data Brokers

- Used in scenario to validate an Identity to mitigate risk
- They collect:
 - Contact Information
 - Identifiers
 - Location
 - Financial Information

Financial Information Data Brokers

- Highly sensitive data used by banks to gauge a person's financial situation and trustworthiness
- Used in conjunction with National Database (like the “Centrale dei rischi”)
- Responsible for one of the worst databreaches in history (Equifax)

The European market is very fragmented and hard to understand, compliance with GDPR changes from nation to nation

Cerved Rating Agency

Cerved Rating Agency is the Italian credit rating agency

Thanks to the work of over 100 dedicated analysts, we contribute to the transparency and integrity of financial markets. Our analyses and views help financial institutions and investors make investment choices and help corporate entities communicate transparently with their stakeholders.

Marketing Data Brokers

- The Previous data brokers used PII directly this type uses so called “inferred data”.
- They Collect a huge number of indirect information and put it together, they use this “anonymized data” to understand the single user
- They connect this info with your advertising ID and cookies to ship you the ads

Since this is all indirect data it cannot be deleted with a data deletion request, but you can reset your cookies and advertising ID.

What about GDPR?

- GDPR is the strongest privacy law in the world, but it is fundamentally reactive.
- Companies use "Dark Patterns", confusing and sometimes illegal cookie banners designed to make you click "Accept All."
- European regulators (like Italy's Garante Della Privacy) are understaffed and underfunded. Each grayzone needs legislature to catch up and properly define it.
- Without it we would be in an awfull place (look at the USA)

What you can do about it

- Level 1: The Easy Swaps (Low Friction):
- Level 2: The Infrastructure Swaps (Medium Friction):
- Level 3: The Hardcore Path (High Friction - Mention only):

Level 1

- Reject cookies
- That's it, it's one of the biggest wins of GDPR and you're blindly accepting everything

I would advise periodically clearing cookies and advertising IDs too





Level 1



Ditch Chrome or Edge or Opera

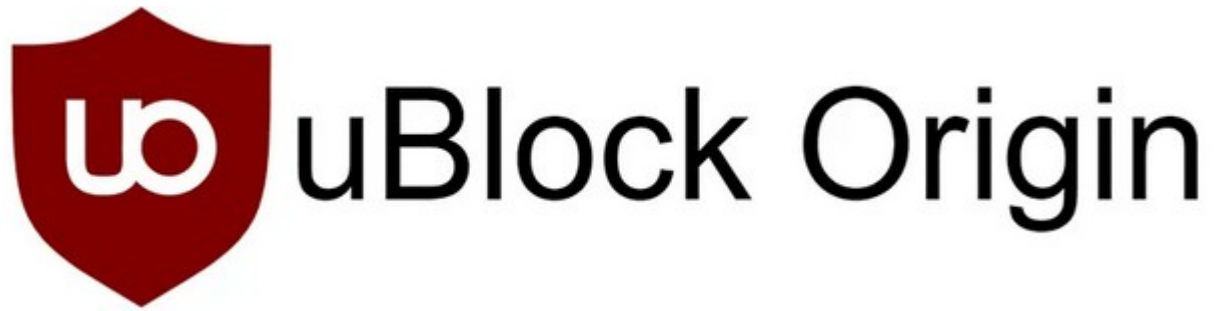
Switch to a browser that respects your privacy:

- Firefox with the telemetry disabled is good but a great options is Librewolf if you don't mind some website functionality being broken
- Brave is okay, but I don't find the company trustworthy, and their built-in ad-blocker is not the best

Pro tip: if you use a browser like librewolf you can use brave or firefox if a website is broken

Level 1

- Install the U-block Origin Ad-blocker
- Remove every browser extension you don't absolutely need, most of them are malware



Level 1

Stop using Google as a search engine

- Use DuckDuckGo (it also has those ai feature if you want them)
- Or another privacy respecting engine(e.g. brave search, SearxNG, startpage, etc.)



DuckDuckGo

Level 1

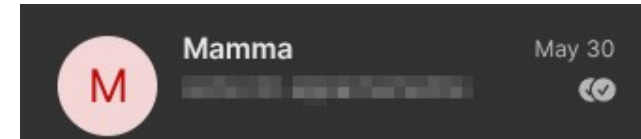


- Don't use SMS or normal Calls
- Use to an end-to-end encrypted service

Okay: Whatsapp (better but it's recent AI integration is terrible and every feature other than text messages is privacy invasive)

Best: Signal, it's a non profit, it has feature parity, best privacy by far, its not hard to make people switch, I now talk to my mom exclusively on signal

P.S:Telegram is not good.



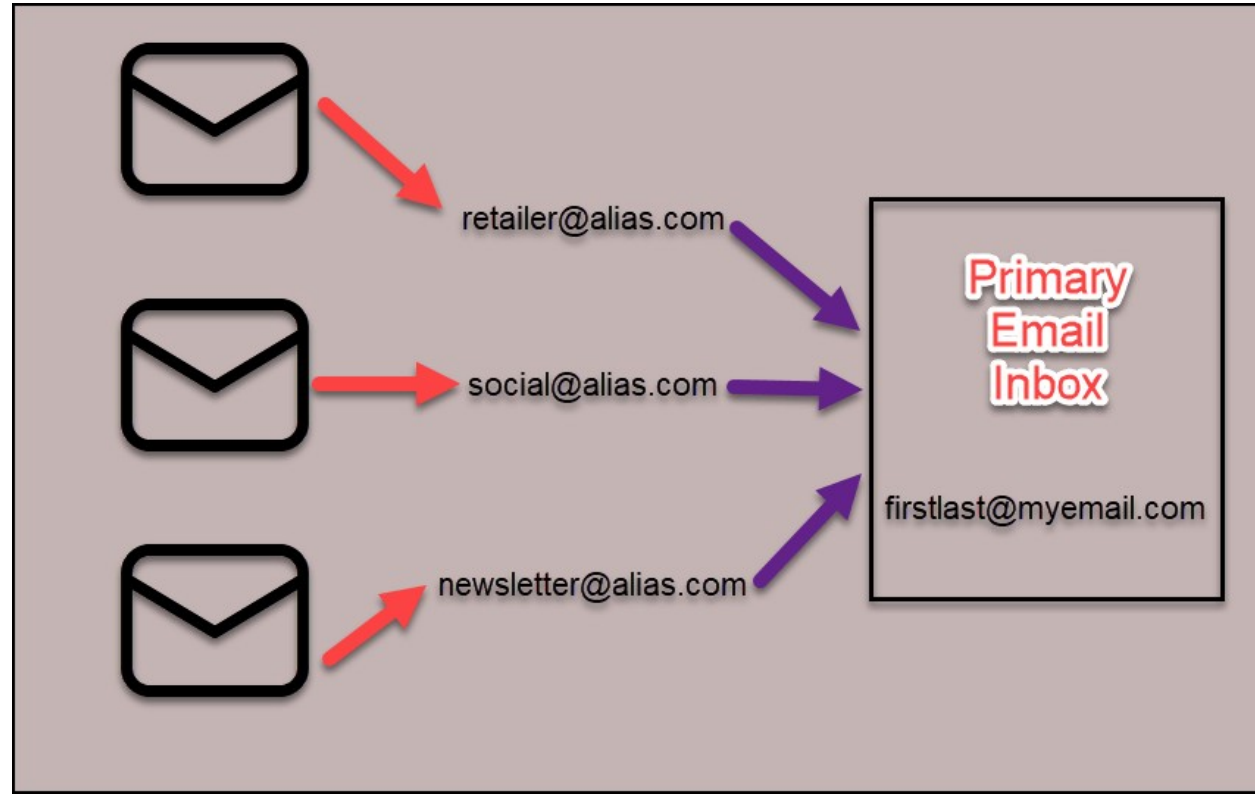
- Change Your Email:

Switch your email from Gmail/Outlook/icloud to Proton or Tutanota

You import your entire email history, setup automatic forwarding to the new email address

Level 2

- Setup Email Aliases



Level 2

- Stop using Google calendar

Both Tutanota and Proton offer calendars in their plans and they allow for easy importing of events.

Level 2

- Use a VPN but don't listen to the marketing it simply hides your IP.

But use the right one :

- Most of the VPNs you can find on the app store are malware
- Never use free VPNs, they are free because they steal your data, if you can't pay for it then you're better off not using a VPN
- My recommendations:
 - ProtonVPN
 - MullvadVPN



Level 2

- Use privacy respecting alternatives

There are many and for almost any product you use, you just need to a little time to find them

My suggestion: Proton's Unlimited plan, it's only 12 euros a month and it has a lot of services

Level 3

- Alternative Front-ends
- Linux
- Custom Android OS (GrapheneOS, CalyxOS)
- Self hosting (immich, nextcloud, home assistant, jellyfin, navidrome, etc)
- Self hosted AI models (Ollama, llama.cpp, Open Web UI)
- Google maps alternatives (OpenStreetMap using CoMaps, Apple Maps)
- AI data poisoning
- Much Much More

Honorable mention

Duck.ai

DuckDuckGo's ai platform

All chats are  private

Ask anything privately



Tools



Fast

GPT-5



Useful Tools

Privacy Visualizer

consumerrights.wiki

alternativeto.net

awesome-privacy.xyz

Products To Avoid

- Ring cameras (most online video surveillance)
- IoT devices (smart fridges, smart mattresses, smart toothbrushes etc)
- Amazon alexa and similar
- Anything Google or Meta(or in general from big tech)
- **Paypal**

Some Harsh Truths:

Social media is really bad

- Your face, your family's face, and your children's face have been fed to AI models

First off, the same principle that goes for the web goes for 99% of apps out there — if you are creating something to post publicly, on a digital platform, chances are it's going into one AI crawler or another, or already has. Remember, most social media apps have, from the beginning, predicated their entire business models on encouraging you to produce content that they will analyze and use to sell you ads with automated systems. Nothing is sacred here, or even truly private, unless the service in question offers end-to-end encryption or particularly good privacy settings.

Instagram/TikTok are inherently toxic to privacy; you either accept it or delete it

Example: I use instagram from my browser and only when I need to

Some Harsh Truths:

Privacy is inconvenient

- Every New website that sell you that shiny new feature that will make your life easier will also do its best to siphon your data away.

Some things can't be fixed, either delete them or accept the consequences

Let's talk AI

The AI Vault Has No Lock

- What goes in, stays in.
- The Myth of the "Average": AI doesn't dissolve your data; it memorizes patterns and can regurgitate them verbatim, private details, proprietary code, and artwork could be extracted 1:1 by other users. [1]
- If you get specific enough with the prompt the training data starts to narrow rapidly

[1] Exploring Memorization and Copyright Violation in Frontier LLMs: A Study of the New York Times v. OpenAI 2023 Lawsuit <https://arxiv.org/html/2412.06370v1>

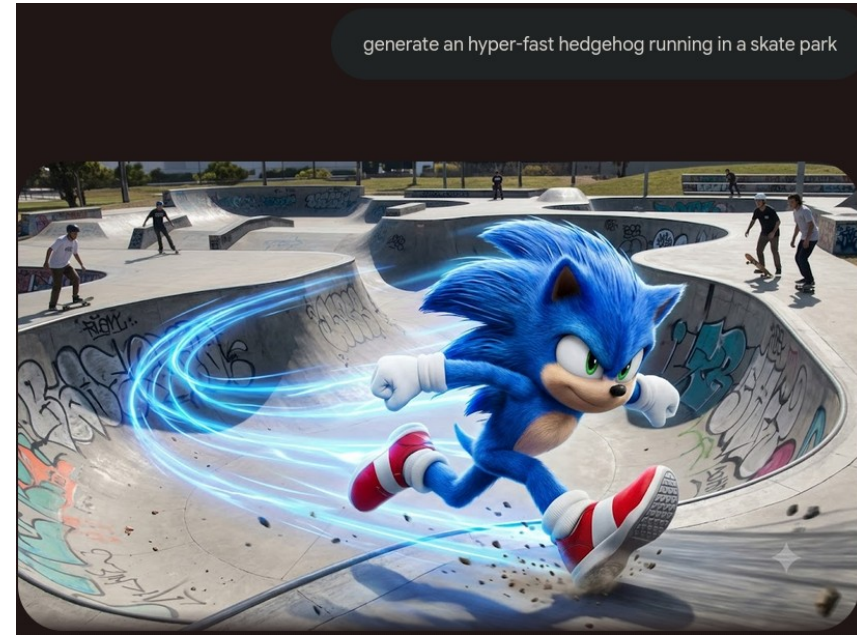
If you wouldn't paste it onto a public billboard, do not type it into an AI prompt.

General prompts pull from millions of sources. Hyper-specific prompts collapse the math down to very few sources.

Note: I have never talked about sonic

In the example its just
a silly image:

With an intricate enough prompt
your personal story could be shared



Data Deletion is practically impossible

- In a traditional database, if you invoke your privacy rights (like GDPR's "Right to be Forgotten"), the company can go into their server and delete your specific row of data. It's gone.
- With AI, it doesn't work that way. Once your data is used to train a model, it is used to adjust millions of invisible mathematical connections (called "weights"). Your data is no longer a separate file; it is baked into the machine's brain.
- For the same reasons that you can't un-bake a cake you also can't remove your data once a model has been trained with it.

Institutional Accounts

I cannot ask you to not use AI.

If you can't use privacy respecting providers
check if you can use an institutional account.
If you use your university account with Gemini it
your chat data *should* not be used for training

Le tue chat di Università di Trento non vengono utilizzate per migliorare i nostri modelli. Gemini è un'AI. Può fare errori, quindi verifica le sue risposte. La tua privacy e Gemini

The future

- In 2026 privacy its not a switch you turn on, its a continuous battle against systems that collect first and ask permission later(if at all).
- identity verification bills.
- There's currently a war on privacy and those that make privacy tools

“The greatest dangers to liberty lurk in insidious encroachment by men of zeal, well meaning but without understanding.”

Call To Action

- Start doing what you can little by little
- Keep yourself Informed.
- Read international press
- Talk to your family and friends.
- Use Free Software
- Give Financial Support to FOSS projects
- Follow political news

Recommended Reads / Videos

- The inevitable fate of your personal data – Reject Convenience (youtube)
- Privacy 101 - Naomi Brockwell TV (youtube)
- Who Will Remember Us When The Servers Go Dark – New Design Congress
- Meredith Whittaker (Signal CEO) interview – Wired
- The Circle – Dave Eggers
- How advertisers ruined a good idea – Reject Convenience (youtube)
- 2030: Privacy's Dead. What happens next? – Tom Scott (youtube)
- Your social media "likes" expose more than you think – Jennifer Golbeck TEDx

This presentation has been made with 100%
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